

Email Spam Classification

Detecting Spam vs Ham using NLP & Machine Learning

Python

NLP

Scikit-learn

Streamlit

Pandas



Project Overview

What We're Building

An end-to-end intelligent system that automatically classifies emails as Spam or Ham (legitimate) using Natural Language Processing and supervised Machine Learning — complete with a Streamlit web app for real-time predictions.



Data
Collection



EDA &
Preprocessing



Model
Training



Streamlit
Deployment

201,105 total emails • 70% Ham • 30% Spam

Dataset Overview

215,140

Total Emails

150,000

Ham Emails (0)

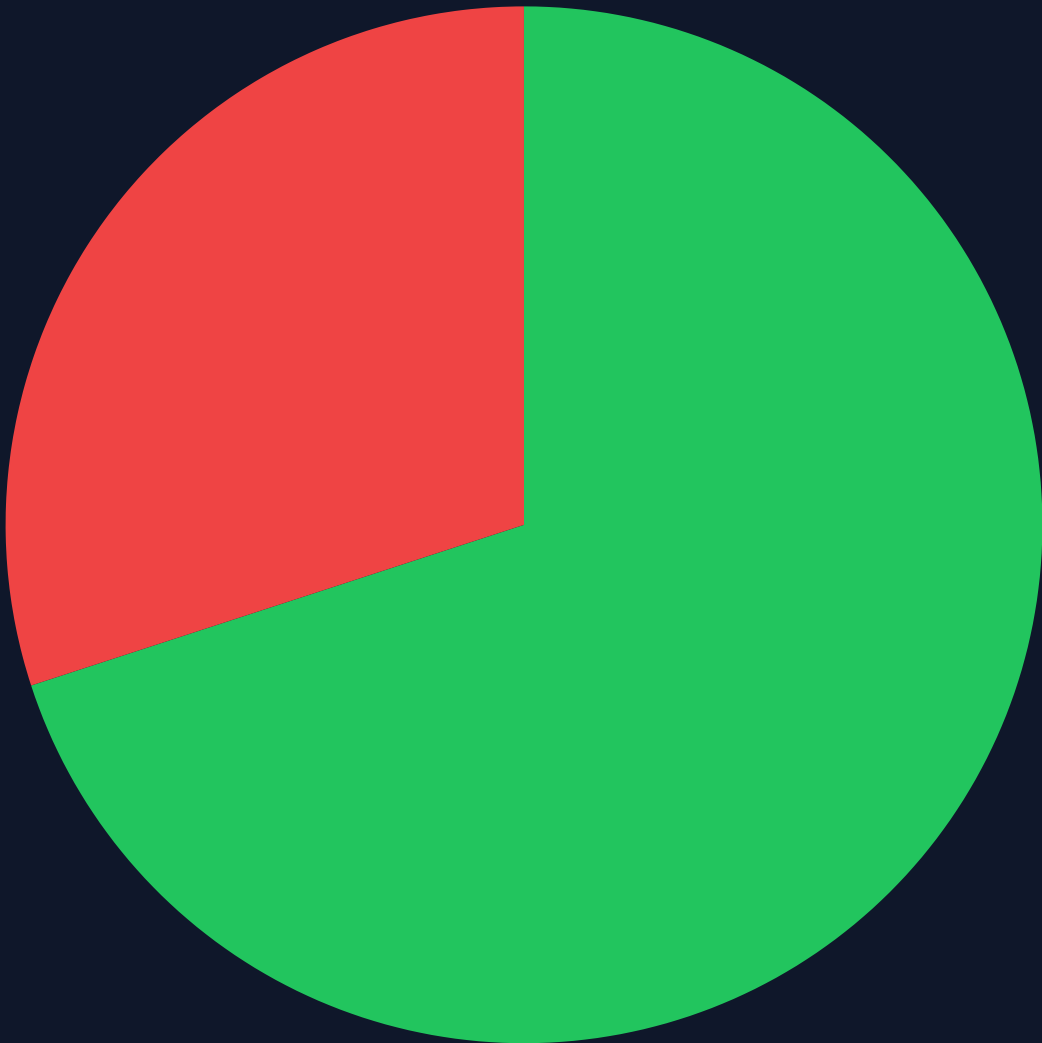
65,000

Spam Emails (1)

3

Columns

● Ham (70%) ● Spam (30%)



Dataset Columns

Unnamed: 0 Row index	int
message Raw email/SMS body text	str
label 0 = Ham, 1 = Spam	object/string before encoding

EDA & Preprocessing Pipeline

Text Cleaning Steps

- 1

Lowercase conversion
- 2

Number removal
- 3

Punctuation removal
- 4

Stopword removal
- 5

Text normalization

EDA Visualizations Created

- 

Spam vs Ham Distribution (countplot)
- 

Word Count Distribution (histplot + KDE)
- 

Character Count Distribution (histplot)
- 

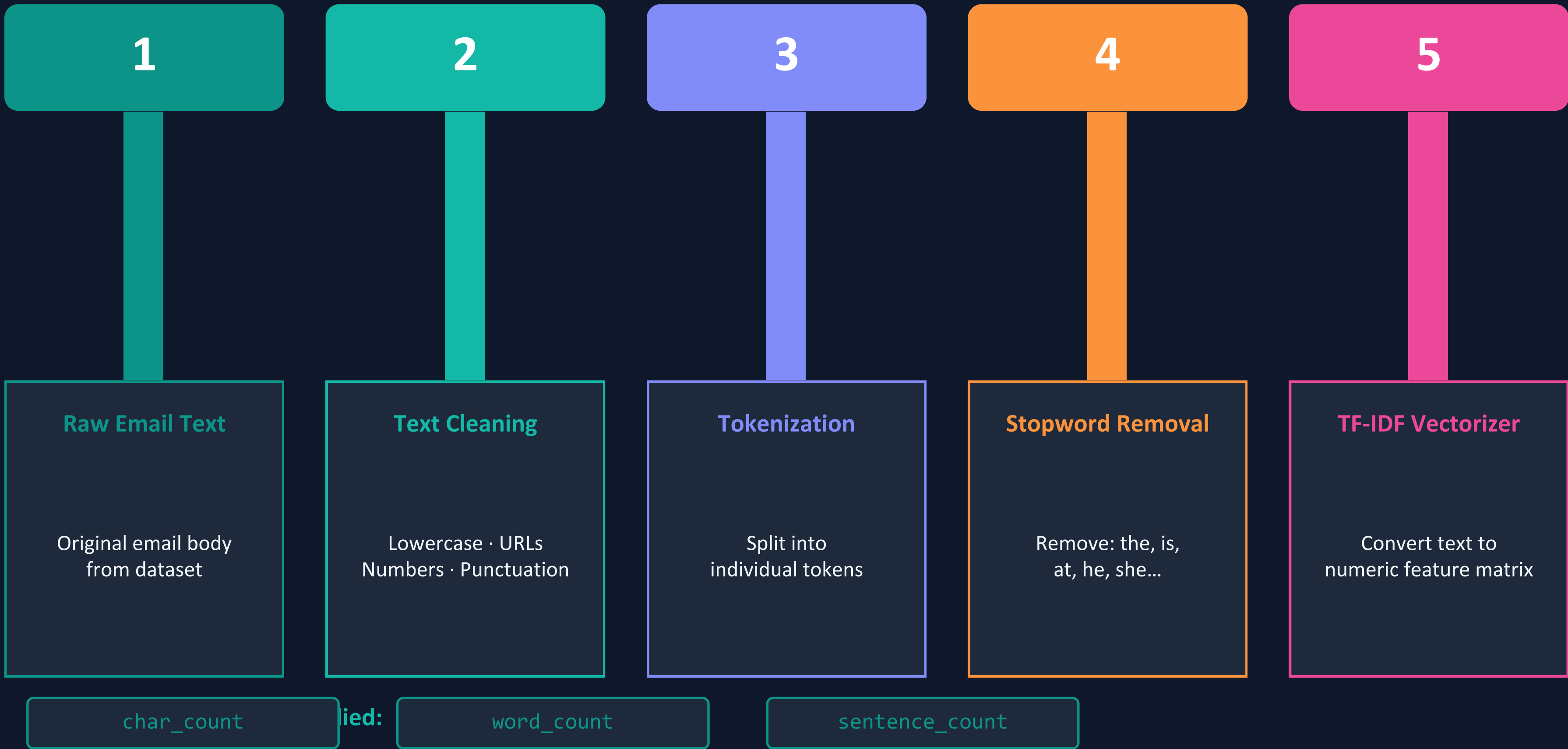
Spam Word Cloud (WordCloud)
- 

Ham Word Cloud (WordCloud)
- 

Top 20 Spam Words (barplot)
- 

“TF-IDF vectorization used to convert cleaned text into numerical feature vectors for machine learning.”
That aligns with your actual notebook and app.

NLP Feature Extraction



Model Training & Evaluation

Algorithms Compared

Algorithm	Accuracy	Precision
Naive Bayes	92.64%	0.870
SVM (Linear Kernel)	94.30%	0.904
LSTM	96.10%	0.95

96.1%

Best Accuracy

0.95%

Best F1 score

LSTM

Best Model



Streamlit Web App — app.py

How the App Works

Load Model

Loads trained SVM spam classification model

Load TF-IDF

Loads tfidf_vectorizer.pkl for transforms

Preprocess Input

Lowercase → digits → punctuation → stopwords

Vectorize

tfidf.transform([cleaned_input])

Predict

model.predict(vector_input) → 0 or 1

Show Result

st.error (Spam) or st.success (Ham)

Core Prediction Logic

```
def preprocess_text(text):  
    text = text.lower()  
    text = re.sub(r'\d+', '', text)  
    # remove punctuation  
    words = text.split()  
    words = [w for w in words  
             if w not in stop_words]  
    return ' '.join(words)  
  
cleaned = preprocess_text(user_input)  
vec = tfidf.transform([cleaned])  
pred = model.predict(vec)  
if pred[0] == 1:  
    st.error('Spam Detected')  
else:  
    st.success('Ham Email')
```

Project Structure & Files

📁 email-spam-classification/

├─ email_dataset_classification.csv

├─ EDA_Preprocessing.ipynb

├─ model_training.ipynb

├─ spam_classifier_model.pkl

├─ tfidf_vectorizer.pkl

├─ app.py

├─ plots.jpg

└─ requirements.txt

File Descriptions

email_dataset_classification.csv

Raw dataset — 201,105 emails, 3 columns

EDA_Preprocessing.ipynb

Data cleaning, EDA plots, feature engineering

Model_Training.ipynb

TF-IDF, model training & evaluation notebook

spam_classifier_model.pkl

Serialized trained ML classifier

tfidf_vectorizer.pkl

Fitted TF-IDF vectorizer for inference

app.py

Streamlit web app for real-time classification



Project Summary

Building smarter inboxes with AI-powered spam detection

- ✓ 215,140 real-world emails
- ✓ End-to-end NLP preprocessing pipeline
- ✓ “Multiple ML and Deep Learning models evaluated
 - LSTM achieved the best performance with 96.1% accuracy.”
- ✓ Deployed as a live Streamlit web app